Venkat Kumar Meda

Full Stack & DevOps Engineer | Python, FastAPI, React | AWS, Kubernetes, GitOps
Visakhapatnam, Andhra Pradesh, India (Willing to relocate)
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Professional Summary

Full Stack & DevOps Engineer with hands-on production experience in building scalable web applications and cloud-native infrastructure. Strong expertise in Python, FastAPI, React, REST APIs, and microservices architecture, along with DevOps practices including CI/CD automation, Docker, Kubernetes (EKS), and AWS cloud deployments. Experienced in monitoring, high-availability systems, infrastructure automation, and AI model deployment pipelines. Proven ability to manage end-to-end software lifecycle from development to production operations.

Education

- **Vignan's Institute of Information Technology** — *Dec 2021 – Mar 2024*
  Bachelor of Technology in Mechanical Engineering — Visakhapatnam, India
  - GPA: 7.8/10

Professional Experience

- **DeepSight AI Labs** — *May 2024 – Present*
  Software Engineer – Computer Vision & DevOps — Gurgaon, India (On-site)
  - Developed enterprise-grade, multi-tier full-stack applications using Python and FastAPI, and built React-based dashboards for real-time monitoring, alerts, and system health visualization, supporting 50+ concurrent users with 99.5% uptime.
  - Migrated 10+ microservices from monolithic deployments to a cloud-native Kubernetes architecture, improving scalability, high availability, and deployment consistency across environments.
  - Designed and implemented a GitHub Actions + ArgoCD GitOps pipeline to automate Docker image builds, SHA-based version tagging, and multi-cluster Kubernetes deployments, reducing release cycle time by 60% and minimizing deployment-related downtime by 95%.
  - Deployed and managed production workloads on AWS (EC2, EKS, RDS, S3) and on-premises Kubernetes clusters, implementing rolling updates, readiness/liveness probes, and automated rollback strategies to ensure near zero-downtime releases.
  - Built secure RESTful APIs using FastAPI with JWT and OAuth2 authentication, integrating MySQL and NoSQL databases with unit testing and documentation, handling 1000+ API requests per minute.
  - Configured monitoring and observability stack (Prometheus, Grafana, CloudWatch) to track system performance, troubleshoot issues, and implement proactive alerting, reducing incident response time by 30%.
  - Collaborated with cross-functional teams to deliver production-grade AI model deployment pipelines for computer vision applications including face recognition and object tracking.

Key Projects

Enterprise Video Surveillance & Real-Time Alert Management Platform (Full Stack / AWS Cloud)
Tools: Python, FastAPI, React JS, WebSockets, SQLAlchemy, MySQL, Docker, Kubernetes, AWS), GitHub Actions, ArgoCD, Git

- Designed and developed an end-to-end enterprise-grade full-stack platform using FastAPI and React, building real-time monitoring dashboards with WebSocket-based alert streaming and system health visualization, serving 50+ concurrent users.
- Developed 20+ secure RESTful APIs using FastAPI with JWT authentication and RBAC, optimizing MySQL queries with SQLAlchemy and indexing to handle 1000+ requests per minute, supported by structured logging and PyTest-based unit/integration testing.
- Containerized services using Docker and deployed workloads on Kubernetes, configuring readiness/liveness probes, persistent storage, resource limits, and rolling update strategies for zero-downtime deployments.
- Designed and implemented a centralized GitOps-based CI/CD pipeline using GitHub Actions and ArgoCD (App-of-Apps pattern), enabling automated Docker image builds, SHA-based version tagging, and synchronized deployments across 6 independent Kubernetes clusters.
- Integrated private Docker Hub registry with secure Kubernetes imagePullSecrets, eliminating manual deployments and reducing human deployment errors by 90%.
- Implemented observability stack using Prometheus, Grafana, and CloudWatch for real-time monitoring, proactive alerting, and faster incident resolution.

Cloud-Native Kubernetes & CI/CD Automation Platform

Tools: Kubernetes, Helm, KEDA, GitHub Actions, Jenkins, Docker, Prometheus, Grafana, Linux

- Migrated 10+ microservices to cloud-native architecture on public cloud and on-premises environments, implementing DevOps automation and release management principles across 40+ Linux nodes.
- Implemented an end-to-end GitHub Actions CI workflow integrated with ArgoCD, enabling automated image versioning, GitOps synchronization, and zero-manual Kubernetes deployments.
- Configured event-driven autoscaling (KEDA) and implemented monitoring solutions (Prometheus, Grafana) for proactive system diagnostics, demonstrating strong troubleshooting and problem-solving skills.
- Applied networking fundamentals to configure service meshes, load balancers, and ingress controllers, ensuring secure communication with encryption and proper IAM policies.
- Collaborated with cross-functional teams to establish DevOps culture, conducting peer code reviews, documentation standards, and knowledge sharing sessions on release management techniques.

Computer Vision AI Pipeline - Production Face Recognition System

Tools: Python, FastAPI, Triton Inference Server, TensorRT, Kubernetes, Docker, Milvus, Linux

- Deployed production AI models for real-time computer vision inference using Python, object-oriented programming, and Web Services architecture, processing 10+ concurrent video streams with 95% accuracy.
- Integrated multiple AI models (YOLOv5, InsightFace, DeepSORT) into enterprise software pipeline with proper unit testing, error handling, and performance optimization on Linux environments.
- Implemented NoSQL vector database (Milvus) for high-performance feature search with sub-second query response times, demonstrating proficiency in database optimization and troubleshooting.
- Optimized inference performance using TensorRT and deployed on Kubernetes with GPU acceleration, applying DevOps practices for scalable AI model serving in production environments.
- Documented technical architecture, API specifications, and deployment procedures, while participating in code reviews to ensure quality and maintainability of the codebase.

TECHNICAL SKILLS

- **Core Roles:** Full Stack Engineer, Backend Engineer, DevOps & Platform Engineer
- **Programming:** Python, SQL, Bash, JavaScript
- **Backend:** FastAPI, REST APIs, Microservices Architecture, API Design, JWT, OAuth2, RBAC
- **Frontend:** React JS, Material-UI, HTML, CSS, WebSockets
- **DevOps & CI/CD:** Docker, Kubernetes, Helm, GitHub Actions, ArgoCD, GitOps, Release Automation
- **Cloud & Infrastructure:** AWS (EC2, EKS, RDS, S3, IAM, VPC, CloudWatch), Linux, NGINX
- **Databases:** MySQL, PostgreSQL, Milvus (Vector DB), SQLAlchemy, Query Optimization
- **Monitoring:** Prometheus, Grafana, CloudWatch, Logging, Performance Tuning
- **Version Control & Practices:** Git, Code Review, Unit Testing (PyTest), Agile Methodology
- **AI & Model Deployment:** OpenCV, YOLOv5, DeepSORT, InsightFace, TensorRT, Triton Inference Server